

Lecture 4:

Classification / Types of Computers

1 Overview: Three Ways to Classify a Computer

The word “computer” covers a huge family of machines — from a wristwatch calculator to a weather-forecasting supercomputer. To study them sensibly, we classify them in three independent ways, depending on what we care about: how they represent data (technology used), how big and powerful they are (size & capacity), and what job they are built to do (purpose). A single real computer can be described using all three labels at once — for example, your laptop is a digital, micro, general-purpose computer.

Classification of Computers		
By Technology Used	By Size & Capacity	By Purpose
Digital Analog Hybrid	Micro Mini Mainframe Super	General Purpose Special Purpose

Figure 1: The classification of computers, redrawn from the lecture slide.

Classification basis	Question it answers	Types covered
By Technology Used	How does it represent and process data?	Digital, Analog, Hybrid
By Size & Capacity	How big, fast and expensive is it, and how many users does it serve?	Micro, Mini, Mainframe, Super
By Purpose	Is it built to do many jobs, or one job?	General Purpose, Special Purpose

2 Classification by Technology Used

According to the technology used to represent and process data, computers are grouped into three types: Digital, Analog, and Hybrid.

2.1 Digital Computers

Digital computers are mainly general-purpose computers that represent and store data as discrete quantities or numbers, rather than as a continuously varying signal.

- All processing is done in terms of the numeric (binary digit) representation of data and information.
- Although the user enters data in decimal or character form, the computer internally converts it into binary digits (0s and 1s).
- Because they work with exact, countable values, digital computers give very precise, repeatable results.

Examples

PC (Personal Computer), Laptop, and virtually every modern general-purpose computer.

2.2 Analog Computers

Analog computers are special-purpose computers that represent and store data using continuously varying physical quantities such as current, voltage, or frequency — rather than discrete numbers.

- They are programmed for measuring physical quantities like pressure, temperature and speed, and for performing computations directly on those measurements.
- Analog computers are mainly used for scientific and engineering applications.

Examples

Thermometer — a simple analog computer that measures temperature. Speedometer — the position of the needle on the dial represents the car's speed.

Limitations of Analog Computers

1) They do not have the logical facilities that programming affords. 2) They cannot store data in large quantities. 3) The cost of implementing computation on them is very high.

2.3 Hybrid Computers

Hybrid computers combine the technology of both analog and digital computers, taking advantage of the strengths of each.

- They store and process analog signals that have been converted into discrete numbers using an analog-to-digital converter.
- They can also convert digital numbers back into analog signals or physical properties using a digital-to-analog converter.
- Hybrid computers are mainly used in artificial intelligence (robotics) and computer-aided manufacturing (e.g., process control).

Examples

Automated Teller Machine (ATM).

2.4 Quick Comparison

Feature	Digital	Analog	Hybrid
Represents data as	Discrete numbers (binary)	Continuous physical quantities	Both — converts between the two
Purpose	General-purpose	Special-purpose	Special-purpose
Precision	Very high, exact	Lower, approximate	High, combines both
Typical use	Everyday computing	Scientific / engineering measurement	AI, robotics, process control
Example	PC, Laptop	Thermometer, Speedometer	ATM

3 Classification by Size and Capacity

This is the most commonly used classification. Moving from micro to super, computers generally get physically larger, faster, more expensive, and able to serve more users at once — though in modern usage a “micro” computer can still be extremely powerful.

3.1 Microcomputers

Microcomputers, also called personal computers (PCs), use a microprocessor as their CPU, along with a memory unit, an input device and an output device.

- They are small in size and do not have large storage capacities.
- They are designed to be used by one person at a time, which is why they are commonly called personal computers.
- Popular uses: word processing, browsing the web, sending/receiving e-mail, spreadsheet calculations, database management, editing photographs, creating graphics, and playing music or games.

Examples

IBM PC, PS/2, Apple II, Macintosh.

3.2 Minicomputers

A minicomputer is a multi-user computer, positioned between a microcomputer and a mainframe in power.

- They have higher processing speed and storage capacity than microcomputers.
- Minicomputers can support 4 to 200 users working simultaneously.
- They are used for real-time applications in industries, research centres, etc.

Examples

PDP-11, IBM 8000 series.

3.3 Mainframe Computers

Mainframe computers are multi-user, multi-programming, high-performance computers.

- They operate at very high speed, have very large storage capacity, and can handle the combined workload of many users at once.
- They are generally used in centralised databases.
- Mainframes are used in organisations like banks or large companies, where many people need frequent access to the same data.

Examples

CDC 6600, IBM ES/9000 series.

3.4 Supercomputers

Supercomputers are the largest and fastest computers in existence.

- A supercomputer has a number of CPUs operating in parallel, which is what makes it so fast.

- They are used for massive data processing and solving highly sophisticated problems — in science and defence, designing and launching missiles, weather forecasting, biomedical research, aircraft design, and automobile design.

Examples

CRAY-3. India has its own series of supercomputers called PARAM, developed by C-DAC, and ANURAG.

3.5 Quick Comparison

Feature	Micro	Mini	Mainframe	Super
Users	Single user	4–200 users	Many users (centralised)	Dedicated, specialised jobs
Relative power	Lowest	Higher than micro	Very high	Highest
Typical setting	Home / personal desk	Industry, research centre	Banks, large companies	Science, defence, weather
Example	IBM PC, Macintosh	PDP-11	CDC 6600, IBM ES/9000	CRAY-3, PARAM

4 Classification by Purpose

By purpose, computers are grouped into two types depending on whether they are built to do many different jobs, or one specific job.

4.1 General Purpose Computers

A general-purpose computer can work on different types of programs given to it, and can therefore be used in countless applications.

- Programs are not permanently stored in the machine; they are input at the time of execution.
- These computers are very versatile.
- Simply by using different software on the same hardware, a general-purpose computer can handle writing and editing (word processing), managing facts in a database, tracking manufacturing inventory, and performing scientific calculations — among many other tasks.

4.2 Special Purpose Computers

A special-purpose computer is designed to perform a specific task.

- The instructions (programs) to carry out that task are permanently stored in the machine.
- For its specific task, this type of computer works very efficiently — but such computers are not versatile.
- Useful applications: graphics-intensive video games, traffic-light control systems, navigational systems in aircraft, weather forecasting systems, and satellite launch / tracking systems.

4.3 Quick Comparison

Feature	General Purpose	Special Purpose
Programs	Input at the time of execution, easily changed	Permanently stored in the machine
Versatility	Very versatile — handles many tasks	Not versatile — handles one task only
Efficiency at its task	Good, but not optimised for any one job	Extremely efficient at its one job
Example	PC running word processor, spreadsheet, browser, etc.	Traffic-light controller, aircraft autopilot, ATM

5 Quick Revision Summary

One page to revise everything above the night before a test.

Basis	Type	One-line definition	Example
Technology	Digital	Represents data as discrete numbers (binary); general-purpose.	PC, Laptop
Technology	Analog	Represents data as continuously varying physical quantities.	Thermometer, Speedometer
Technology	Hybrid	Combines analog and digital using A-to-D and D-to-A converters.	ATM
Size	Micro	Personal computer built around a microprocessor; single user.	IBM PC, Macintosh
Size	Mini	Multi-user computer supporting 4–200 users; used for real-time work.	PDP-11
Size	Mainframe	Multi-user, multi-programming, high-performance; centralised database.	CDC 6600, IBM ES/9000
Size	Super	Largest, fastest; many CPUs working in parallel.	CRAY-3, PARAM
Purpose	General Purpose	Runs many different programs, loaded at execution time.	A typical PC
Purpose	Special Purpose	Permanently programmed for exactly one task; not versatile.	ATM, traffic-signal control

Memory hook

Technology asks “how does it store data?” (Digital / Analog / Hybrid). Size asks “how big and how many users?” (Micro < Mini < Mainframe < Super). Purpose asks “one job or many?” (General / Special).

6 Practice Questions

6.1 One-Word / Fill-in-the-Blank

- The _____ converter changes an analog signal into digital form inside a hybrid computer.
- A computer built for one permanent task, such as an ATM, is called a _____ purpose computer.
- Minicomputers can support between _____ and _____ users simultaneously.
- India's series of supercomputers, developed by C-DAC, is called _____.
- A thermometer is a simple example of a(n) _____ computer.

6.2 Short Answer Questions (2–3 marks each)

- Define a digital computer and give two examples.
- List the three limitations of analog computers.
- What is the difference between a general-purpose and a special-purpose computer?
- Name any two examples each of minicomputers and mainframe computers.
- Why are supercomputers able to work so much faster than other computers?

6.3 Long Answer / Essay Questions (7–10 marks each)

- Explain, with a labelled diagram, the classification of computers by technology used. Discuss digital, analog and hybrid computers in detail, with examples.
- Describe the classification of computers by size and capacity. Compare microcomputers, minicomputers, mainframe computers and supercomputers on the basis of users, speed, storage and typical applications.
- “A special-purpose computer is efficient but not versatile, while a general-purpose computer is versatile but not specialised.” Explain this statement with suitable examples.

6.4 Answer Key — One-Word Questions

#	Answer
1	Analog-to-digital (A/D) converter
2	Special
3	4 and 200
4	PARAM
5	Analog